The Environment Tower: An Identity-Free Neural Encoder of Hockey Context

OptBot Phase A — design rationale, specification, and honest limits

OptBot Research
companion to `ARCHITECTURE_PLAN.md` v2.2 · regenerable from `scripts/21-25`

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Abstract

A hockey player's observed statistics blend who he is with what surrounded him: linemates, opponents, coach, score state, zone, fatigue, rink. Any system that projects a player onto a *new* team must separate the two, because the question itself is "keep the player, swap the surroundings." Phase A is a small ($\sim$2M parameter) neural encoder with one job: compress everything surrounding a player during a stretch of ice time into a single vector h_{env} while **provably knowing nothing about the player himself**. We motivate the identity-free constraint against two rejected alternatives, specify every input and every exclusion with its measured evidence, describe the training-time failure modes and the probe gates that catch them, derive the double-counting obligation (K-1) that the tower imposes on the player-side model, and state candidly where the tower serves — and where it is *not required by* — the October 2026 proof event.

1 The problem, stated precisely

For player p, destination environment E', and decision time t_0, we seek

$$\theta(p, E', t_0) \;=\; \mathbb{E}\big[\,Y_{1:40} \,\big|\, \mathrm{do}(E \sim E'),\; \mathcal{H}_p(<t_0)\,\big]\,, \tag{1}$$

the expected on-ice impact over the first $\sim$40 games if the environment is *set* to E', conditioned only on information knowable before t_0. The operator $\mathrm{do}(\cdot)$ is the entire company. Associational models learn that players in good environments produce more; a trade decision needs the effect of *placing* this player into that environment. A model can be excellent at prediction and useless for the intervention.

For (1) to be computable, the model must factor into independently swappable pieces:

$$\widehat{Y} \;=\; g\big(\,h_{\text{env}}(E),\; \tau(p),\; \ell\,\big), \tag{2}$$

with h_{env} the environment representation (Phase A), $\tau(p)$ the player's talent channel (Phase C), and ℓ the deployment levers. Phase B is g. This factorization is not an aesthetic; it is the handle the intervention pulls.

2 The central design decision: identity-freeness

The most important sentence in this paper: **the tower never sees the player it is describing the ice for** — no ID, no statistics, no priors, no deployment history. Three designs were considered.

Rejected: the monolith. One network eating player + environment + usage. It would predict well and answer nothing: with all factors entangled, "swap the team" has no handle — the internal representation of the player is partly a representation of his linemates and his team's system, and the environment effect is confounded by every player who selected into that environment.

Rejected: the firewalled two-stream (our own prior design). The legacy Phase A (still on disk as `encoder.py`; deprecated July 2026) placed a focal-identity stream and a context stream in one module, the context side reading a gradient-detached identity summary through a one-way attention "firewall." The audit verdict: such firewalls are *promises, not proofs*. Gradient paths can be blocked; information paths through shared batch statistics and co-adapted training dynamics cannot be certified, and every leak is invisible until it surfaces as a confidently wrong answer on exactly the rows we sell — movers. The cheapest firewall is absence.

Chosen: absence, plus a separate frozen talent channel. The tower's inputs are surroundings only; the player enters once, in Phase B, through the pre-computed channel $\tau(p)$ into which no gradient flows. "Swap the environment" becomes a literal code path: recompute h_{env} with the destination's people and context; $\tau(p)$ does not move.

> **The residual threat: roster fingerprinting.** Five teammate IDs nearly identify the team, the line, and often the focal player (who else centers Draisaitl and Hyman?). A tower that learns "this WITH-set ⇒ elite production" has partially memorized the focal player without seeing his ID. Active defenses: (i) teammate dropout 20% — the WITH-set is never reliable enough to fingerprint; (ii) probe P3-incremental every epoch — a ridge probe asks whether h_{env} predicts focal talent *better than raw teammate-quality features alone*; if yes, the checkpoint is rejected; (iii) the K-1 re-derivation (§7) reabsorbs whatever linemate credit the tower legitimately claims. Priced, probed, gated — not assumed away.

3 Inputs: every seat earned, every ban evidenced

Governing law, enforced at import by `contracts/tower_schema.py`: every input is SYNTH (exactly constructible for a hypothetical destination from pre-t_0 data), MARG (servable as a distribution via the slot bootstrap), or BANNED. The test is serving-time honesty: could we produce this number on trade day, for a trade that has not happened?

3.1 The people (the intervention itself)

- **Member identities** (`with_ids`, `vs_ids`, ≤ 5 per side): learned 96-d vectors from a deliberately starved shared table ($\sim$1,700 players). Who is beside and against him is the largest environmental fact and the one the v0 baseline cannot see.

- **Exposure** (`with_seconds`, `vs_seconds`): how much of each human; drives the pooling bias (§4.2).
- **Dynamic shells** (per member, as of game date): age, position, handedness, *season-lagged* recent form, and the member's own games-since-team-change. The shell distinguishes 2019-Kane from 2025-Kane under one ID vector. Form is season-lagged by construction (the F-1 echo rule): form earned *next to* the focal player reflects the focal player back into h_{env} — identity by reflection. Measured: naive-current form correlates 0.310 with focal talent; lagged form 0.267; the survivor is genuine assortative line-building the tower *should* see.

3.2 The situation, the bench, the schedule

Score bucket, period and time bucket; zone start, faceoff location, start regime, entry mode, stoppage class, after-icing (with OZ/DZ splits; on-the-fly entries measured $\sim$12% better than DZ faceoffs; zone information decays with $\sim$20 s half-life — probed, §5.3); home/away, rink id, bench rights, long change, skater differential. Team priors (pace, xG rates, EB-shrunk goalie tier, both teams; all strictly pre-game and lag-audited); coach id + tenure (82 coaches, shrinkage; a new coach is UNK with tenure 0, itself a signal); rest days and back-to-backs (audited: 13.6% b2b rate); focal `games_since_team_change` as an *explicit* input (F-2), so the settling curve is modeled rather than smeared; entry-side timing features (decisions made before the exposure).

3.3 Banned, with evidence

Banned	**Reason (measured where applicable)**
the focal player, in any form	§2; the core discipline
`duration` as input	windows end at whistles; whistles correlate with events; stop-bias +51%. Survives only as a loss weight
exit-side timing	players leave *because* of what happened; outcome-caused $\Rightarrow$ label peeking. Entry-side twin stays legal — the asymmetry is the point
in-window event counts	outcomes wearing feature costumes
season / era ids	undefined at serving; era enters honestly via drifting team priors
matchup-quality %, standings prior	traced to leaky legacy computations (realized full-game overlap weights; unverifiable as-of branch)
focal deployment priors	they describe the *old* coach's choices; the destination role template is a lever, not an environment fact

4 Architecture

4.1 Member representation: static core ⊕ dynamic shell

$$m_j = \mathrm{MLP}\big(\,e_{\mathrm{id}(j)} \oplus s_j\,\big), \qquad s_j = [\,\text{age},\ \text{pos},\ \text{hand},\ \text{lagged form},\ \text{member gsc}\,]. \tag{3}$$

Identity vectors carry what is stable; shells carry what changes. The split is also the **UNK mode**: an unseen player (an AHL call-up today; an entire AHL roster in the future recruiting product) receives the UNK vector plus a real shell — graceful degradation from "I know this man" to "I know his type."

4.2 Pooling: seconds-biased attention

$$a_j \propto \exp\left(q^\top m_j + \log \sec_j\right), \qquad h_{\text{with}} = \textstyle\sum_j a_j\, m_j. \tag{4}$$

Not a mean (a 3-second cameo would equal a full shift), not a max (one member dominates), not plain attention (over-focus; resurrects fingerprinting). The log-seconds bias makes exposure the prior that attention adjusts around. Padding (id 0, seconds 0) is masked exactly; audited at 100% consistency over 200k sampled windows.

4.3 Context, fusion, and size

$$h_{\text{env}} = \text{Fusion}([\, h_{\text{with}},\, h_{\text{vs}},\, h_{\text{ctx}}\,] + \text{CLS}) \in \mathbb{R}^{256}, \tag{5}$$

with two pre-norm attention blocks: context must *modulate* how the people matter (the same five opponents mean different things protecting a lead versus chasing). Total ∼2M parameters, and small is a feature: 6.5M windows sounds large, but each identity vector learns only from that player's appearances. A large tower memorizes windows; a starved one is forced to learn hockey. The GBDT twin (§6) referees exactly this.

5 Training, and every trap we know about

5.1 No loss of its own

The tower trains end-to-end with the Phase B trunk against window outcomes: "environment quality" is *defined* as whatever representation makes the trunk right across two million windows. The price is that Phase B pathologies can reach the tower through the gradient — hence the plumbing below.

5.2 The weighting stack (who gets a vote)

Mechanism	**Prevents**
loss × seconds (exposure weight)	low-minute windows dominating: a 6-second rate observation gets ∼1/100 the vote of a 10-minute stretch
heteroscedastic heads (predict μ, σ)	noisy windows steering: Gaussian NLL divides error by σ^2, auto-muting short exposures
switch-weighting ×3 post-move	the regime we sell drowning in the settled majority; safe because gsc is an explicit input
`is_multistint` downweight	blended-row noise
EB shrinkage on $\tau(p)$ (Phase C)	a 40-career-minute player carrying an extreme talent value anywhere

5.3 The probe scorecard (every epoch; red rejects the checkpoint)

Probe	Question	Red means
P3-incremental	does h_{env} predict focal talent beyond raw teammate-quality features?	roster fingerprint; widen dropout, starve IDs
zone-decay	does the zone-start effect decay at the measured $\sim$20 s half-life?	start effects smeared over windows
era	can h_{env} identify the season beyond team priors?	time travel at serving
coach	does the coach embedding explain variance beyond roster?	(want *some*) scheme vs. roster proxy
UNK-degradation	replace members with UNK+shell: graceful drop?	tower depends on memorized identity; AHL mode dead

Era-split checkpoints ($\leq$2021/22/23/24); all ablations era-fixed. The 2018–19 season currently trains shell-only for member form (no prior season on the surface until the 2017–18 build lands); if that build stalls, the season is excluded rather than trained on a hole.

6 The joint with Phase B

$$x = [\, h_{\text{env}} \oplus \tau(p) \oplus \text{bio}\,] \;\rightarrow\; 4 \times 768 \text{ SwiGLU residual trunk, FiLM}(\ell_{1:17}) \;\rightarrow\; 10 \text{ heads.} \tag{6}$$

Three pipes, physically separate. $\tau(p)$ is frozen (no gradient); the 17 deployment levers enter via FiLM modulation — they *reshape* the computation rather than add to it, which is what later answers "what if we used him differently?" without retraining. **Heads:** MoneyPuck xG for/against as primary (heteroscedastic Gaussian) with rush/rebound/chaos splits; in-house xG as auxiliary witness; counts as Poisson with exposure offsets; Kendall–Gal weighting across heads. **The legitimate firewall:** micro-stat heads (hits/giveaways/takeaways) are gradient-detached from the tower — arena scorers disagree by up to 2.25× on giveaways (T19), and that noise may not sculpt h_{env}. Unlike §2's rejected firewall, a gradient detach is a one-line mechanical guarantee against a non-adversarial threat: different threat class, different confidence. **Two gates, two questions:** the tower must beat the slot-average environment in era-fixed ablation (do specific humans help?); the trunk must beat a GBDT twin on the same tabular inputs (is the neural machinery earning its keep?).

7 The joint with Phase C

7.1 Talent as a residual

$$\tau(p) = \text{EBshrink}\left(\sum_{w \in \mathcal{W}_p} \lambda^{\Delta t_w} \left[\, y_w - \mu_{\text{base}}(E_w)\,\right] \right), \tag{7}$$

decay-weighted, league-demeaned, out-of-fold (the baseline never saw the windows it judges), shrunk toward zero with fitted K in effective minutes. This channel is the engine of the already-proven 7.6% harness win.

7.2 The double-counting obligation (K-1)

If the baseline μ_{base} is identity-blind, then $\tau(p)$ contains credit that belongs to p's linemates. Upgrading the environment model to one that *does* credit linemates (the tower) while keeping the old residuals counts the same value twice:

$$\underbrace{h_{\text{env}}\text{ claims Draisaitl}}_{\text{new}} + \underbrace{\tau(\text{McDavid-adjacent})\text{ still contains Draisaitl}}_{\text{old}} \Rightarrow \text{ systematic overprediction for stars-adja} \tag{8}$$

In v0 the two errors partially cancel; the tower breaks the cancellation — and the failure lands on exactly the GM question ("is he real without 97?"). Hence hard ship-gate K-1: before v1 ships, talent is re-derived as the residual against the *tower's* environment (one EM round), and the top/bottom-15 sniff must re-rank sensibly. Non-negotiable.

7.3 The v2 frontier (not in the MVP)

A style vector (rush sniper / net-front / point bomber; 13,708 pre-computed shot-personality cards) interacted with the tower's people-vectors becomes the chemistry engine — gated behind an identifiability census, because chemistry claims without co-occurrence support are astrology.

8 Serving: from hypothetical to number

Query: player → destination → role. The scenario builder bootstraps ×2000 over the destination slot's *real* pre-t_0 windows, weighted by the joint future schedule (opponent, venue, rest — never independent marginals; no Frankenstein windows), integrates games 1..40 of the settling curve via gsc, and emits projected on-ice xGF% with a conformal band whose coverage was *measured* at 80.0% against an 80% claim on 860 historical moves — binned by evidence volume × intended role delta (K-2: a third-liner queried as a first-liner draws his band from the wide-change error pool). Outside joint support: `extrapolated: true`; far outside: `REFUSED_OOS`. Refusal is a feature; it is what makes the other answers credible.

9 MVP alignment — and honest disalignment

The MVP bar (July 2026): proof sufficient for an NHL team partnership or seed funding: the engine (v0 beats the industry baseline by 7.6%, CI clear), sealed September predictions, the public scoreboard, and the market-calls log (gated by its own backtest). One target, one event, one raise.

Where the tower serves the goal

(i) The upgrade path with a floor: v1 ships only if it beats v0 on the same 860-move harness, CI-clear, all seven gates; October cannot be hurt by the attempt. (ii) The demo moment: v0 cannot answer "and with Kane instead?" — the tower can; interactivity is what makes the demo an engine rather than a spreadsheet. (iii) The differentiation claim "first causal simulation engine in sports" requires do(·)-capable machinery; (2) is that machinery, defensible in diligence. (iv) UNK/shell mode is the AHL recruiting product's front door — the expansion pitch shares the tower.

Where it does not serve the goal (read this too)

(i) October does not require it; if GPU access slips past mid-August, the correct call is to seal v0 and keep training. (ii) The market-calls leg deliberately does not use it (shot cards + role minutes + v0 environment; fewer moving parts under deadline). (iii) It charges the K-1 tax — real re-validation days v0 never costs. (iv) Its risk is real: the honest prior is a modest first-try harness gain over v0, because the talent channel — which both share — carries most of the signal. The gates exist because "neural = better" is a hypothesis, not an entitlement.

10 Questions the reader should ask (the grill list)

1. *If the tower never sees the player, how is the projection about him?* — Phase B injects him via $\tau(p)$; the tower prices the ice for a generic player, talent shifts it to him. Chemistry is v2; ask what deferring it costs.
2. *Five teammate IDs nearly name him — why isn't that fatal?* — dropout + P3-incremental + K-1; demand the probe numbers on the first real checkpoint.
3. *Why allow teammate form at all?* — season-lagged only; the surviving correlation is measured assortativity; exact pairwise leave-focal-out is v1.1.
4. *What if the tower fails?* — v0 ships; the company never depends on the net.
5. *Why believe the bands?* — measured 80.0% on 860 real moves, per-cell after K-2; extrapolations flagged, not disguised.
6. *Weakest input?* — serving-time opponent lineups (team known, five sampled); a standing sensitivity ablation owns it.
7. *Does any of this delay October?* — no; GPU is the only gating resource, and v0 is the floor.

Status ledger (July 22, 2026)

Training surface: BUILT — 6.49M windows, 7 seasons, member shells aligned; lag-audit FLAGGED 0; deep audit 15 checks ALL CLEAR ×3 seasons. Feature legality: enforced in code. Tower+trunk v2.2: specified here; `scripts/19` implements next (the on-disk `encoder.py` is the deprecated two-stream — do not train it). GPU: **the only blocker**. Ship gates: seven, in `ARCHITECTURE_PLAN.md` §5.

Internal references: `ARCHITECTURE_PLAN.md` v2.2 · `PHASE_A_FEATURES.md` · `AUDITOR_GUIDE.html` · scripts 21–25. Every number in this paper regenerates from the repository.